

Web Technology (IT 4th and Soft 2nd)

(Final Tutorial)

(Deadline: 2082-04-15)

1. Why do we need hosting? Explain domain name registration process?.
2. What are web servers and web browsers? Outline the steps involved in registering a domain name, and explain the hierarchy of a domain name system.
3. Justify “web architecture helps to understand web application”. Also, explain working principle of HTTPs protocol.
4. Describe the fundamental standards of web architecture. Also, explain why the HTTP protocol is considered stateless, multimedia-independent, and a connection-oriented protocol.
5. Write HTML tags to generate the following design:

Roll no:	Name	Subjects			
		C	Web	Math	C++
1	Abhisek	50	45	60	70
2	Suresh	50	45	45	NQ
3	Yodhin	80	85	90	75
Contact to department if Not Qualified(NQ)					

6. Design Following Layout:

Day/Time	7:00 - 8:40	8:40 - 10:20	10:20 - 11:10	11:10 - 12:50	12:50 - 2:30
Sunday	Web	DBMS	Break	Algebra	Microprocessor
Monday	OS	Algebra		Web	Microprocessor
Tuesday	Web	DBMS		OS	Microprocessor
Wednesday	Microprocessor	DBMS		Web	OS
Thursday	Web	DBMS		Algebra	Microprocessor
Friday	Web	DBMS		OS	Microprocessor
Above Routine , effective from coming sunday.					

7. Importance of CSS in web development? Define the purpose of the following CSS properties: text-transform, z-index, scale, and outline.
8. Explain with example, the relationship between display, justify-content and align-items of CSS properties. Additionally, write html example with pseudo elements and pseudo classes.
9. What is the role of selectors in CSS? Provide examples of the: blur, :focus, and :hover pseudo-classes, as well as the ::before and ::after pseudo-elements.
10. Why CSS is used in Web designing? Explain different levels of styles sheets with example. Also write advantages and disadvantages of each.
11. Create an HTML form that allows users to input their Full Name, Email, and Mobile Number. Then, write a JavaScript program to validate this user data and display a success message if the inputs are valid.
12. Write a JavaScript program which validates the user data (name, email, contact no) and displays the success message if validated successfully. Also create the HTML form for entering the data.
13. Perform the following array methods and explain what each one does.
 - i. join()
 - ii. toString()

- iii. pop()
- iv. push()
- v. shift()
- vi. unshift()
- vii. slice()
- viii. splice()

14. Define dynamic content. Write JavaScript code to illustrate stacking elements and locating mouse cursor.
15. What is meant by the DOM (Document Object Model)? Write a JavaScript program that dynamically appends elements to an unordered list () based on the values stored in a JavaScript array. Additionally, represent DOM hierarchy of above program.
16. Define Asynchronous in JS. How can you write an asynchronous JavaScript program that reads data from a students.json file and present each record in HTML table?
17. Create a JavaScript program that retrieves student details from a student.json file using XMLHttpRequest and displays the records in an HTML table format.
18. How php actually works, along with HTML, CSS and JS? Write php script with associative array and show key-value as following:

Person Information

Key	Value
Name	Harish Chaudhary
Age	30
Email	harish@gmail.com
City	Kathmandu
Profession	Web Developer

19. GET vs POST. Write content to “mycontent.txt” file on Submission.

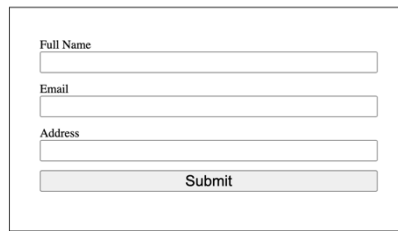
Write your content

20. Explain types of file modes. Write text area content in “content.txt” file. Also validate content not more than 200 words, if exist alert “Limit your content”.

Write your content

21. Why Regular Expression is used ? How does PHP support pattern matching ? Explain with example.
22. Describe Starting , Storing and Destroying a Session. Explain with sample codes. Also , differentiate between Session and Cookies.
23. Explain how a session is initiated, stored, and destroyed in PHP. Also, discuss the purpose of using sessions and cookies.

24. Write a PHP script that handles user login(Email and Password). If the login is successful, redirect the user to homepage.php; otherwise, display a “Sorry” message.
25. Write PHP script to insert form data to database table on form submit. Where every data must be in lowercase before inserting into table.



The image shows a simple web form with a rectangular border. Inside the border, there are three text input fields stacked vertically. The first field is labeled 'Full Name', the second 'Email', and the third 'Address'. Below these fields is a single button labeled 'Submit'.

26. Short Notes
- a) let, const and var keywords
 - b) Rest and Spread Parameter
 - c) Image mapping
 - d) Hyper Links and Multimedia in HTML
 - e) Php array sorting methods
 - f) GLOBAL variables in PHP